MATH50003 Numerical Analysis

V.2 Discrete Fourier Transform

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Part V

Approximation Theory

1. Fourier Expansions and approximating Fourier coefficients
2. Discrete Fourier Transforms and interpolation

V.2.1 The discrete Fourier transform

Map from values to approximate Fourier coefficients

Definition 35 (DFT)**.** The *Discrete Fourier Transform (DFT)* is defined as:

$$
\begin{aligned}
Q_n &:= \frac{1}{\sqrt{n}} \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \mathrm{e}^{-\mathrm{i}\theta_1} & \mathrm{e}^{-\mathrm{i}\theta_2} & \cdots & \mathrm{e}^{-\mathrm{i}\theta_{n-1}} \\ 1 & \mathrm{e}^{-\mathrm{i}2\theta_1} & \mathrm{e}^{-\mathrm{i}2\theta_2} & \cdots & \mathrm{e}^{-\mathrm{i}2\theta_{n-1}} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \mathrm{e}^{-\mathrm{i}(n-1)\theta_1} & \mathrm{e}^{-\mathrm{i}(n-1)\theta_2} & \cdots & \mathrm{e}^{-\mathrm{i}(n-1)\theta_{n-1}} \end{bmatrix} \\
&= \frac{1}{\sqrt{n}} \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega^{-1} & \omega^{-2} & \cdots & \omega^{-(n-1)} \\ 1 & \omega^{-2} & \omega^{-4} & \cdots & \omega^{-2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{-(n-1)} & \omega^{-2(n-1)} & \cdots & \omega^{-(n-1)^2} \end{bmatrix}
\end{aligned}
$$

for the n-th root of unity $\omega = \mathrm{e}^{2\pi\mathrm{i}/n}$.

Proposition 1 (DFT is Unitary) $Q_n \in U(n)$, that is, $Q_n^\star Q_n = Q_n Q_n^\star = I$.

Example 21 (Computing Sum)**.**

V.2.2 Interpolation

Approximate Fourier series interpolates at sample points

Corollary 5 (Interpolation).

$$f_n(\theta) := \sum_{k=0}^{n-1} \hat{f}_k^n \mathrm{e}^{\mathrm{i}k\theta}$$

interpolates f *at* θ_j*:*

$$f_n(\theta_j) = f(\theta_j)$$

Example 22 (DFT versus Lagrange)**.**